

## Signal Processing

### Exercise 10 – IIR, Correlation, Linear / Circular Convolution – Solution

SoSe 2026

#### IIR Filter Design

##### Question 1

An IIR low-pass filter is required to have the following specifications:

- Pass-band edge frequency  $\omega_{\text{pass}} = 0.24\pi$  rad/sample
- Stop-band edge frequency  $\omega_{\text{stop}} = 0.5\pi$  rad/sample
- Maximum pass-band attenuation  $\delta_{\text{pass}} \leq 1$  dB
- Minimum stop-band attenuation  $\delta_{\text{stop}} \geq 4$  dB

Consider a butterworth low-pass analog filter to approximate the filter characteristic. Use the bilinear transform with  $T = 2$  to design the filter. In addition, we would like to make sure that the maximum passband attenuation is met exactly.

The magnitude-squared approximation of a low-pass  $N$ th-order Butterworth filter reads as

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}. \quad (1)$$

or

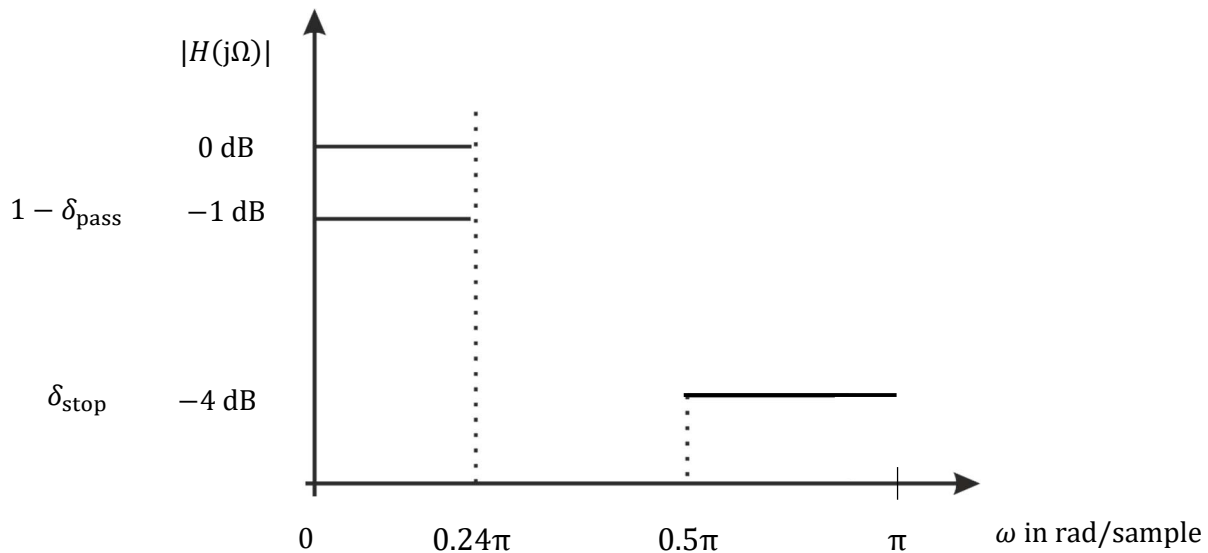
$$|H_c(s)|^2 = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}} \quad (2)$$

Thus, the order of filter is given by

$$N \geq \frac{1}{2} \frac{\log\left(\frac{1}{|H_c(j\Omega_{\text{pass}})|^2 - 1}\right) - \log\left(\frac{1}{|H_c(j\Omega_{\text{stop}})|^2 - 1}\right)}{\log \Omega_{\text{pass}} - \log \Omega_{\text{stop}}}. \quad (3)$$

Why IIR filter? We can get the same approximation accuracy (of the magnitude response) as FIR, but with a lower filter order. The tradeoff is a linear phase.

a) Sketch the tolerance scheme of the filter.



b) Calculate the minimum order  $N$  as well as the cut-off frequency in such a way that the filter specifications are fulfilled.

### Filter order $N$

$$|H_c(j\Omega_{\text{pass}})|^2 = (1 - \delta_{\text{pass}})^2 \quad \text{and} \quad |H_c(j\Omega_{\text{stop}})|^2 = \delta_{\text{stop}}^2$$

This leads to

$$N \geq \frac{1}{2} \frac{\log\left(\frac{1}{(1 - \delta_{\text{pass}})^2} - 1\right) - \log\left(\frac{1}{\delta_{\text{stop}}^2} - 1\right)}{\log \Omega_{\text{pass}} - \log \Omega_{\text{stop}}} .$$

with  $1 - \delta_{\text{pass}} = -1 \text{ dB} = 0.8919$  and  $\delta_{\text{stop}} = -4 \text{ dB} = 0.631$  .

Now consider  $\Omega = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$  and  $T = 2$ .

Nonlinear relationship between the analog frequency and the digital frequency → **frequency warping**

$$\rightarrow \Omega_{\text{pass}} = \frac{2}{T} \tan\left(\frac{\omega_{\text{pass}}}{2}\right) = 0.3959$$

$$\rightarrow \Omega_{\text{stop}} = \frac{2}{T} \tan\left(\frac{\omega_{\text{stop}}}{2}\right) = 1$$

$$N \geq \frac{1}{2} \frac{\log\left(\frac{1}{0.8919^2} - 1\right) - \log\left(\frac{1}{0.631^2} - 1\right)}{\log(0.3959) - \log(1)} = 0.9559$$

We chose  $N = 1$ .

### Cut-off frequency

Take Eq. 1 into account:  $(1 - \delta_{\text{pass}})^2 = \frac{1}{1 + \left(\frac{\Omega_{\text{pass}}}{\Omega_c}\right)^{2N}}$

$$\frac{1}{(1 - \delta_{\text{pass}})^2} = 1 + \left(\frac{\Omega_{\text{pass}}}{\Omega_c}\right)^{2N}$$

Solve for  $\Omega_c$

$$\rightarrow \Omega_c = \frac{\Omega_{\text{pass}}}{\sqrt{\frac{1}{(1 - \delta_{\text{pass}})^2} - 1}} = 0.7808$$

- c) Determine a digital filter system function  $H(z)$  by using the bilinear transform with  $T = 2$ . Hint: Determine  $|H_c(s)|^2$  and  $H_c(s)$  first. In order to do so, try to obtain the poles of  $|H_c(s)|^2$ . Calculate the filter coefficients  $b_k$ ;  $a_k$  of the standardized system function as well.

In order to design our filter, the first step involves solving for the poles by setting the denominator  $1 + \left(\frac{s}{j\Omega_c}\right)^{2N}$  to zero. This will allow us to solve the equation for the Butterworth filter more easily.

Therefore, we obtain

$$s = j \Omega_c \sqrt[2N]{-1}.$$

Taking the relationships  $-1 = e^{j(\pi+2\pi k)}$  and  $j = e^{\frac{j\pi}{2}}$  into account, our equation takes the form

$$s_k = \Omega_c e^{j\left(\frac{2\pi}{2N}k + \frac{\pi}{2N} + \frac{\pi}{2}\right)}, \text{ for } k = 0, \dots, 2N - 1.$$

The solutions (poles) are uniformly distributed in the complex  $s$ -plane around a circle with its radius  $\Omega_c$ . Inserting  $N = 1$  and  $k = 0 ; 1$  gives two poles

$$s_0 = -\Omega_c ; \quad s_1 = \Omega_c.$$

By making use of the identity  $|H_c(s)|^2 = H_c(s) \cdot H_c(-s)$ , and by considering the stable poles on the left half  $s$ -plane to  $H_c(s)$ , we obtain

$$|H_c(s)|^2 = \frac{\Omega_c}{s + \Omega_c} \cdot \frac{\Omega_c}{\Omega_c - s}.$$

Opting for the stable pole (left half of the  $s$ -plane), we get

$$H_c(s) = \frac{\Omega_c}{s + \Omega_c} = \frac{0.7782}{s + 0.7782}$$

Apply the bilinear transform:  $s = \frac{2}{T} \cdot \frac{z-1}{z+1} = \frac{2}{T} \cdot \frac{1-z^{-1}}{1+z^{-1}}$

$$H_c(z) = \frac{\Omega_c}{\frac{1-z^{-1}}{1+z^{-1}} + \Omega_c} = \frac{\Omega_c \cdot (1+z^{-1})}{1-z^{-1} + \Omega_c \cdot (1+z^{-1})} = \frac{\Omega_c + \Omega_c z^{-1}}{1-z^{-1} + \Omega_c + \Omega_c z^{-1}}$$

### Calculation of coefficients $b_k$ ; $a_k$

Standardized form of the system function  $H(z)$ :

$$H(z) = \frac{\sum_{k=0}^N b_k z^{-k}}{1 + \sum_{k=1}^M a_k z^{-k}}$$

Rewrite standardized system function  $H(z)$ :

$$H(z) = \frac{b_0 + b_1 z^{-1}}{1 + a_1 z^{-1}}$$

Rewrite system function  $H_c(z)$ :

$$H_c(z) = \frac{\Omega_c + \Omega_c z^{-1}}{1 - z^{-1} + \Omega_c + \Omega_c z^{-1}} = \frac{\Omega_c + \Omega_c z^{-1}}{(1 + \Omega_c) + (\Omega_c - 1)z^{-1}} = \frac{\frac{\Omega_c}{1 + \Omega_c} + \frac{\Omega_c}{1 + \Omega_c} z^{-1}}{1 + \frac{\Omega_c - 1}{1 + \Omega_c} z^{-1}}$$

By comparing the coefficients, we obtain

$$b_0 = b_1 = \frac{\Omega_c}{1 + \Omega_c} = 0.4385$$

$$a_1 = \frac{\Omega_c - 1}{1 + \Omega_c} = -0.1231$$

IIR filter design – general steps

1. Determine desired frequency response in terms of analog frequencies.
2. Convert the desired frequencies into digital frequencies.
3. Map those digital frequencies into analog frequencies.
4. Design an analog filter for those frequencies.
5. Transform analog filter into digital filter using the bilinear transformation.



## Cross-correlation

When we correlate a given signal with another signal to find out how much resemblance exists between them, then this correlation type is called cross-correlation.

$$\varphi_{xy}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot y[n+m] = \sum_{n=-\infty}^{\infty} x[n-m] \cdot y[n]$$

The maximum of the cross-correlated signal  $\varphi_{xy}[m]$  occurs, when both signals overlap at their best. In this case, the two considered signals become most similar to each other.

Application: Determine signal delays (Radar, Sonar, acoustical distance measurement)

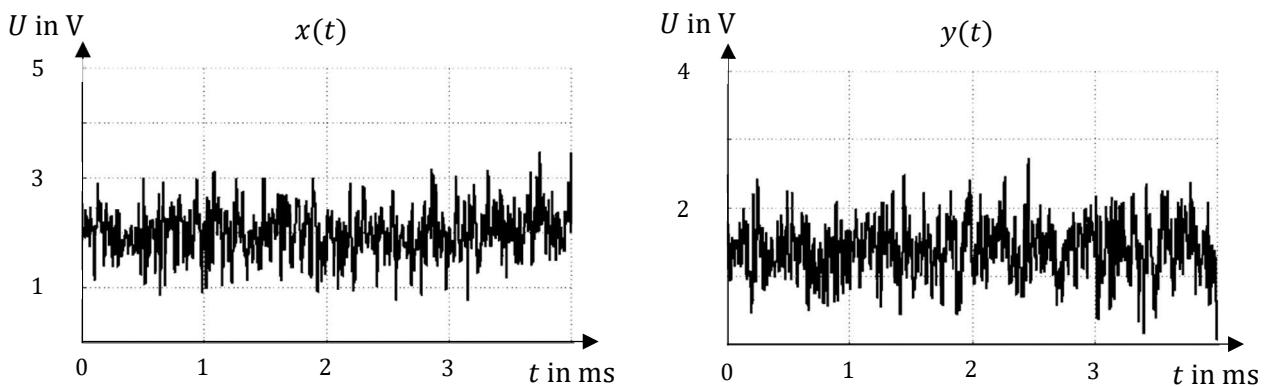


Fig.1: Transmitted signal  $x(t)$  and received signal  $y(t)$ .

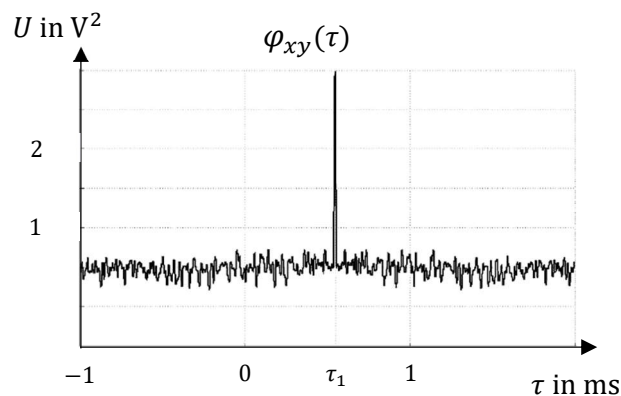
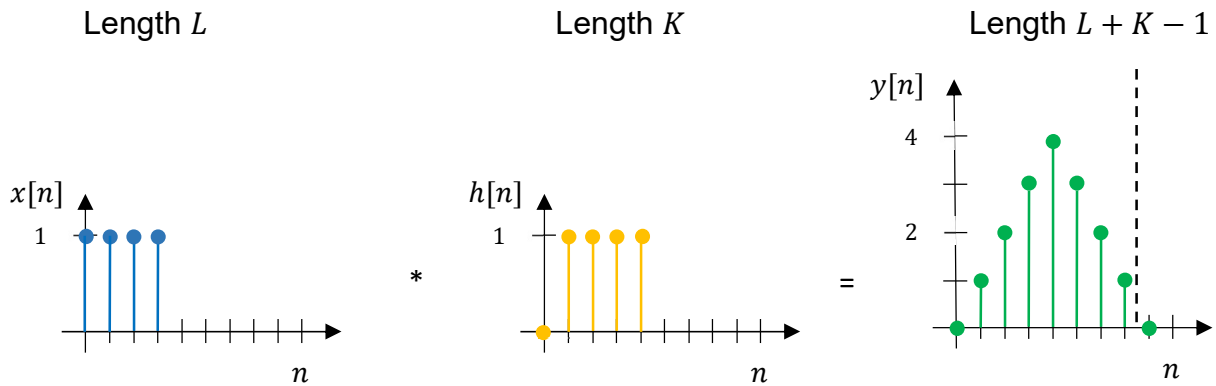


Fig.2: Cross-correlation function  $\varphi_{xy}(\tau)$  as a result of cross-correlating  $x(t)$  with  $y(t)$ .

As depicted in Fig.2, white noise is a very good test signal for time delay measurements using cross-correlation, because there is only one cross-correlation peak at  $\tau_1$ .

## Linear Convolution:

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{+\infty} x[m] \cdot h[n - m]$$



When both signals are congruent (100 % overlapping), then we get a maximum value for  $y[n]$ .

## Circular Convolution:

Circular convolution is applied for periodic sequences, whereas linear convolution is employed for aperiodic sequences.

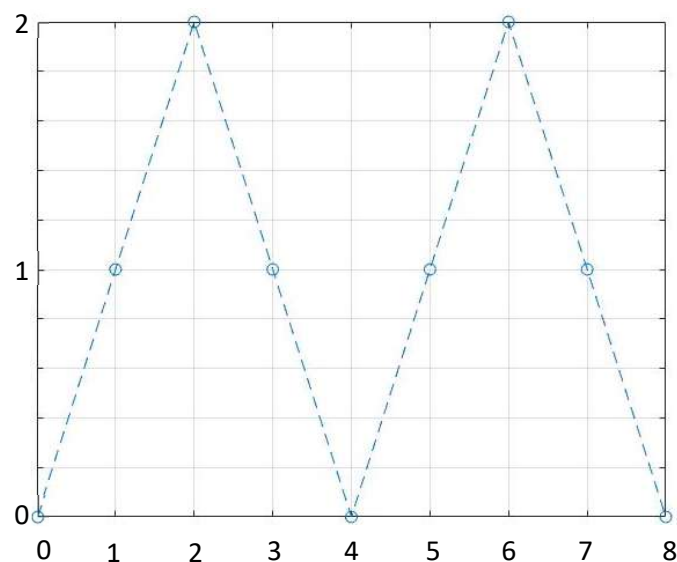
$$y[n] = x[n] \circledast h[n] = \sum_{i=0}^{N-1} x[i] \cdot h[n - i]_N$$

## Question 1

The autocorrelation is applied to determine a certain frequency component of an audio signal. In considering the occurring maxima of the autocorrelation  $\varphi_{xx}[m]$  one can determine the frequency.

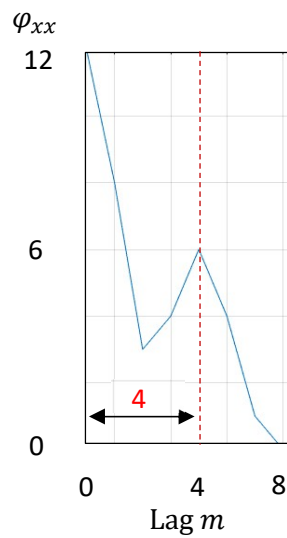
Determine the frequency of the given signal  $x[n] = [0, 1, 2, 1, 0, 1, 2, 1, 0]$  by calculating the autocorrelation function  $\varphi_{xx}[m]$ .

$y(t)$  or  $y[n]$  respectively in V



Time  $t$  in s or sample values  $n$  respectively

| $m$ | Signal values |   |   |   |   |   |   |   |   | $\varphi_{xx}[m]$ |
|-----|---------------|---|---|---|---|---|---|---|---|-------------------|
|     | 0             | 1 | 2 | 1 | 0 | 1 | 2 | 1 | 0 | X                 |
| 0   | 0             | 1 | 2 | 1 | 0 | 1 | 2 | 1 | 0 | 12                |
| 1   |               | 0 | 1 | 2 | 1 | 0 | 1 | 2 | 1 | 8                 |
| 2   |               |   | 0 | 1 | 2 | 1 | 0 | 1 | 2 | 3                 |
| 3   |               |   |   | 0 | 1 | 2 | 1 | 0 | 1 | 4                 |
| 4   |               |   |   |   | 0 | 1 | 2 | 1 | 0 | 6                 |
| 5   |               |   |   |   |   | 0 | 1 | 2 | 1 | 4                 |
| 6   |               |   |   |   |   |   | 0 | 1 | 2 | 1                 |
| 7   |               |   |   |   |   |   |   | 0 | 1 | 0                 |
| 8   |               |   |   |   |   |   |   |   | 0 | 0                 |



$$f = \frac{1}{4 \cdot 1} \text{ Hz} = \frac{1}{4} \text{ Hz}$$

## Question 2

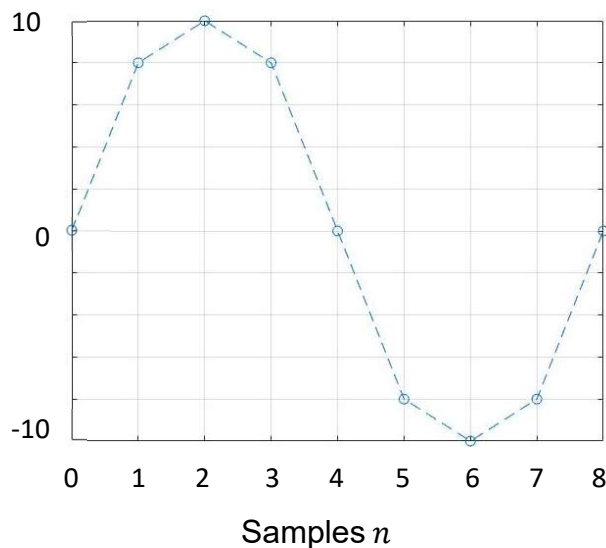
You want to determine the distance from a sound source to an object. Thus, you decide to transmit a single pulse signal. The signal is sampled at  $f_s = 10$  kHz, the experiments are conducted in water (sound velocity  $c_W = 1500 \frac{\text{m}}{\text{s}}$ ).

Apply the cross-correlation and calculate the distance  $d$  from the transmitter to the object.

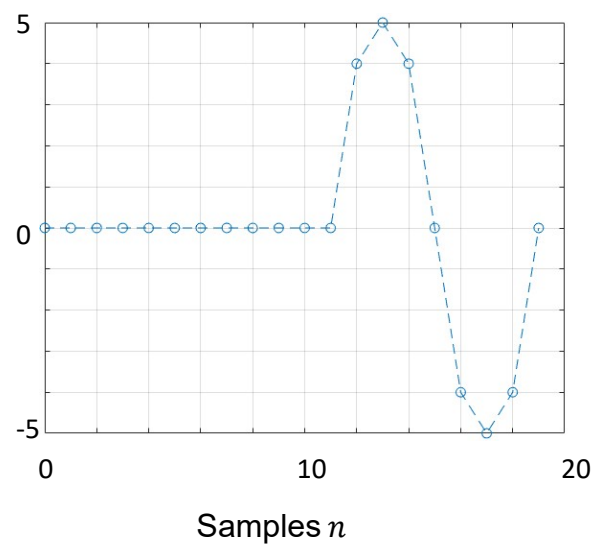
Transmitted signal:  $x[n] = [0, 8, 10, 8, 0, -8, -10, -8, 0]$

Received signal:  $y[n] = [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 4, 5, 4, 0, -4, -5, -4, 0]$

$x[n]$  in V



$y[n]$  in V



| $m$ | Signal values |     |     |     |     |     |     |     |     | $\varphi_{xy}[m]$ |
|-----|---------------|-----|-----|-----|-----|-----|-----|-----|-----|-------------------|
|     | 0             | 4   | 5   | 4   | 0   | -4  | -5  | -4  | 0   | X                 |
| 3   | 0             |     |     |     |     |     |     |     |     | 0                 |
| 4   | -8            | 0   |     |     |     |     |     |     |     | 0                 |
| 5   | -10           | -8  | 0   |     |     |     |     |     |     | -32               |
| 6   | -8            | -10 | -8  | 0   |     |     |     |     |     | -80               |
| 7   | 0             | -8  | -10 | -8  | 0   |     |     |     |     | -114              |
| 8   | 8             | 0   | -8  | -10 | -8  | 0   |     |     |     | -80               |
| 9   | 10            | 8   | 0   | -8  | -10 | -8  | 0   |     |     | 32                |
| 10  | 8             | 10  | 8   | 0   | -8  | -10 | -8  | 0   |     | 160               |
| 11  | 0             | 8   | 10  | 8   | 0   | -8  | -10 | -8  | 0   | 228               |
| 12  | 0             | 0   | 8   | 10  | 8   | 0   | -8  | -10 | -8  | 160               |
| 13  | 0             | 0   | 0   | 8   | 10  | 8   | 0   | -8  | -10 | 32                |
|     |               |     |     |     |     |     |     |     |     |                   |
|     |               |     |     |     |     |     |     |     |     |                   |
|     |               |     |     |     |     |     |     |     |     |                   |

$$\varphi_{xy}[11] = 228 = \varphi_{\max}$$

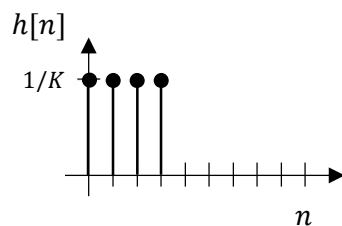
Time difference  $m = 11$  Samples.  $s = \frac{t \cdot c}{2} = \frac{\frac{1}{10000} \frac{s}{\text{sample}} \cdot 11 \text{ samples} \cdot 1500 \frac{m}{s}}{2} = 0.825 \text{ m.}$

### Question 3

Your experiments generate a noisy signal  $x[n] = [1, -3, 2, -1, 2]$  of the length  $L = 5$  and you decide to smooth the signal by using a moving average filter (3<sup>rd</sup> order).

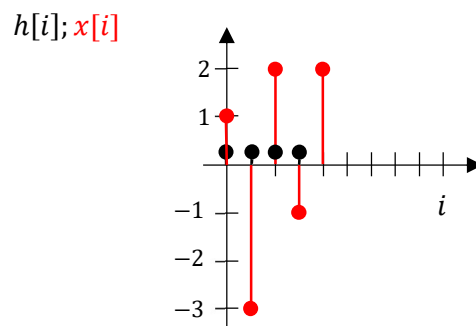
a) Draw the impulse response  $h[n]$ .

3<sup>rd</sup> order  $\rightarrow$  length  $K = 4$

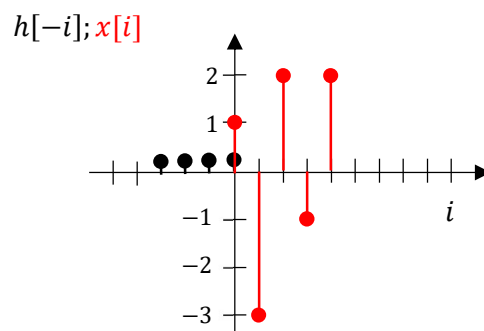


b) Apply the linear convolution to calculate the output signal  $y[n]$ .

1. Plot both signals



2. Mirror impulse response:  $h[i] \rightarrow h[-i]$



3. Shift impulse response  $h[-i]$   $n$ -positions to the right ( $n = 0, 1, \dots, L + K - 1$ ).  
Perform shift by setting  $i = i - n$ .
4. Multiply and sum.

|          | $i \rightarrow$ |      |      | 1    | -3   | 2    | -1   | 2    |      |      |      |  |                |           |
|----------|-----------------|------|------|------|------|------|------|------|------|------|------|--|----------------|-----------|
| $h[0-i]$ | 0.25            | 0.25 | 0.25 | 0.25 |      |      |      |      |      |      |      |  | $y[0] = 0,25$  | $\approx$ |
| $h[1-i]$ |                 | 0.25 | 0.25 | 0.25 | 0.25 |      |      |      |      |      |      |  | $y[1] = -0,5$  |           |
| $h[2-i]$ |                 |      | 0.25 | 0.25 | 0.25 | 0.25 |      |      |      |      |      |  | $y[2] = 0$     |           |
| $h[3-i]$ |                 |      |      | 0.25 | 0.25 | 0.25 | 0.25 |      |      |      |      |  | $y[3] = -0,25$ |           |
| $h[4-i]$ |                 |      |      |      | 0.25 | 0.25 | 0.25 | 0.25 |      |      |      |  | $y[4] = 0$     |           |
| $h[5-i]$ |                 |      |      |      |      | 0.25 | 0.25 | 0.25 | 0.25 |      |      |  | $y[5] = 0,75$  |           |
| $h[6-i]$ |                 |      |      |      |      |      | 0.25 | 0.25 | 0.25 | 0.25 |      |  | $y[6] = 0,25$  |           |
| $h[7-i]$ |                 |      |      |      |      |      |      | 0.25 | 0.25 | 0.25 | 0.25 |  | $y[7] = 0,5$   |           |

#### Question 4

Calculate the cyclic convolution of the two series  $x[n] = [1, 2, 3, 2, 3, 4]$  and  $h[n] = [1, -1, 2, -2, 1, 0]$  for  $N = 6$ .

|                |   | 1  | 2  | 3  | 2  | 3  | 4  | $y[n]$ |
|----------------|---|----|----|----|----|----|----|--------|
| $\leftarrow n$ | 0 | 1  | 0  | 1  | -2 | 2  | -1 | 2      |
|                | 1 | -1 | 1  | 0  | 1  | -2 | 2  | 5      |
|                | 2 | 2  | -1 | 1  | 0  | 1  | -2 | -2     |
|                | 3 | -2 | 2  | -1 | 1  | 0  | 1  | 5      |
|                | 4 | 1  | -2 | 2  | -1 | 1  | 0  | 4      |
|                | 5 | 0  | 1  | -2 | 2  | -1 | 1  | 1      |